OAVS TGT COMPUTER SCIENCE

EXAM PREPARATION SERIES

TOPIC 6

Escape Sequences in C

`\n \t \b \r \a \\ \" \' \0`

150 MCQs with Explanations & Practice Tests

Output-Based | Concept | Advanced | Rapid Fire

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150	6	11	100%
MCQs	Sections	Escape Sequences	Syllabus Coverage

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QUICK REFERENCE: ESCAPE SEQUENCES IN C

Escape Seq	Name	ASCII	Description
`\n`	Newline	10	Moves cursor to next line
`\t`	Horizontal Tab	9	Moves cursor to next tab stop
`\b`	Backspace	8	Moves cursor one position back
`\r`	Carriage Return	13	Moves cursor to start of current line
`\a`	Alert/Bell	7	Produces an audible beep
`\\`	Backslash	92	Prints a literal backslash
`\"`	Double Quote	34	Prints a literal double quote
`\'`	Single Quote	39	Prints a literal single quote
`\0`	Null Character	0	String terminator
`\f`	Form Feed	12	Advances to next page
`\v`	Vertical Tab	11	Vertical tab spacing
`\?`	Question Mark	63	Prevents trigraph interpretation
`\xhh`	Hex Escape	varies	Character by hex ASCII value
`\ooo`	Octal Escape	varies	Character by octal ASCII value

SECTION 1: FUNDAMENTALS OF ESCAPE SEQUENCES

1. What is an escape sequence in C?

A) A sequence of alphabets

B) A special character combination starting with backslash

C) A loop construct

D) A preprocessor directive

Answer: (B)

Explanation: An escape sequence in C is a combination of characters that starts with a backslash (\) and represents a special/non-printable character.

2. Which character initiates an escape sequence in C?

A) /

B) @

C) \

D) #

Answer: (C)

Explanation: The backslash character (\) is used to start an escape sequence in C.

3. What does \n represent in C?

A) Null character

B) New line

C) Next tab

D) Non-printable

Answer: (B)

Explanation: \n is the newline escape sequence that moves the cursor to the beginning of the next line.

4. What does \t represent in C?

A) Tab (horizontal)

B) Top margin

C) Text cursor

D) Termination

Answer: (A)

Explanation: \t represents a horizontal tab and moves the cursor to the next tab stop (usually 8 spaces).

5. What does \b represent?

A) Bold text

B) Backspace

C) Beginning of line

D) Bell sound twice

Answer: (B)

Explanation: \b is the backspace escape sequence. It moves the cursor one position back (left).

6. What does \r represent in C?

A) Return key sound

B) Carriage return

C) Recursive call

D) Right shift

Answer: (B)

Explanation: \r is a carriage return. It moves the cursor to the beginning of the current line without advancing to the next line.

7. What does \a represent in C?

A) Addition operator

B) Alarm/Bell character

C) Array pointer

D) ASCII value

Answer: (B)

Explanation: \a is the alert/bell escape sequence. It produces an audible beep (on systems that support it).

8. What does \\ represent in C?

A) Two backslashes

B) A single backslash character

C) A division operator

D) Escape prefix

Answer: (B)

Explanation: To print a single backslash, you use \\ in C. The first backslash is the escape character, and the second is the literal character.

9. What does \" represent in C?

A) Two double quotes

B) A literal double quote character

C) String delimiter

D) Comment start

Answer: (B)

Explanation: \" prints a literal double quote character inside a string without terminating it.

10. What does \' represent in C?

A) Start of character constant

B) Literal single quote

C) Apostrophe comment

D) String terminator

Answer: (B)

Explanation: \' represents a literal single quote character inside a character constant.

11. What is the ASCII value of the newline character \n?

A) 8

B) 9

C) 10

D) 13

Answer: (C)

Explanation: The newline character \n has ASCII value 10.

12. What is the ASCII value of the tab character \t?

A) 7

B) 8

C) 9

D) 10

Answer: (C)

Explanation: The horizontal tab \t has ASCII value 9.

13. What is the ASCII value of \b (backspace)?

A) 7

B) 8

C) 9

D) 10

Answer: (B)

Explanation: The backspace character \b has ASCII value 8.

14. What is the ASCII value of \a (bell)?

A) 5

B) 6

C) 7

D) 8

Answer: (C)

Explanation: The alert/bell character \a has ASCII value 7.

15. What is the ASCII value of \r (carriage return)?

A) 10

B) 11

C) 12

D) 13

Answer: (D)

Explanation: The carriage return \r has ASCII value 13.

16. Which escape sequence represents the null character?

A) \n

B) \0

C) \null

D) \z

Answer: (B)

Explanation: \0 represents the null character (ASCII value 0), used to terminate strings in C.

17. Which escape sequence is used for a form feed?

A) \f

B) \ff

C) \fd

D) \fm

Answer: (A)

Explanation: \f is the form feed escape sequence (ASCII 12), used to move to the top of the next page.

18. Which escape sequence represents a vertical tab?

A) \h

B) \vt

C) \v

D) \tab

Answer: (C)

Explanation: \v represents the vertical tab character (ASCII 11).

19. How many characters does an escape sequence occupy in memory?

A) Two characters

B) One character

C) Zero characters

D) Depends on type

Answer: (B)

Explanation: Although an escape sequence is written with two (or more) characters, it represents and occupies only ONE byte in memory.

20. Which of the following is NOT a valid escape sequence in C?

A) \n

B) \t

C) \p

D) \r

Answer: (C)

Explanation: \p is not a defined escape sequence in C. The valid ones include \n, \t, \r, \b, \a, \\, \', \", \0, \f, \v.

SECTION 2: OUTPUT-BASED QUESTIONS

21. What is the output of: printf("Hello\nWorld");

```
printf("Hello\nWorld");
```

A) Hello\nWorld

B) Hello World

C) Hello World

D) Compilation error

Answer: (C)

Explanation: \n moves the cursor to the next line. Output:
Hello
World

22. What is the output of: printf("A\tB\tC");

```
printf("A\tB\tC");
```

A) ABC

B) A B C

C) A B C

D) A\tB\tC

Answer: (C)

Explanation: \t inserts a horizontal tab between A, B, and C.

23. What is the output of: printf("Hello\bWorld");

```
printf("Hello\bWorld");
```

A) HelloWorld

B) Hello World

C) HellWorld

D) Hello\bWorld

Answer: (C)

Explanation: \b moves cursor one step back, so 'o' of Hello is overwritten by 'W' of World → HellWorld.

24. What is the output of: printf("Hello\rWorld");

```
printf("Hello\rWorld");
```

A) HelloWorld

B) WorldHello

C) World

D) Hello

Answer: (C)

Explanation: \r moves cursor to the beginning of the current line. 'World' overwrites 'Hello', giving 'World'.

25. What is the output of: printf("1\n2\n3");

```
printf("1\n2\n3");
```

A) 1 2 3

B) 1\n2\n3

C) 1 2 3

D) 123

Answer: (C)

Explanation: Each \n moves to a new line. Output is 1, 2, 3 on separate lines.

26. What is the output of: printf("\t\tOK");

```
printf("\t\tOK");
```

A) OK

B) OK (with two tabs)

C) \t\tOK

D) Error

Answer: (B)

Explanation: Two \t characters insert two horizontal tabs before OK.

27. What will printf("\"Hello\""); print?

```
printf("\"Hello\"");
```

A) \"Hello\"

B) "Hello"

C) Hello

D) Compilation error

Answer: (B)

Explanation: \" prints a literal double quote. So the output is: "Hello"

28. What is the output of: printf("C:\\Programs\\");

```
printf("C:\\Programs\\");
```

A) C:\\Programs\\

B) C:\Programs\

C) C:Programs

D) Error

Answer: (B)

Explanation: \\ prints a single backslash. So C:\\Programs\\ → C:\Programs\

29. What is the output of: printf("ab\bcd");

```
printf("ab\bcd");
```

A) abcd

B) acd

C) abcd (with backspace)

D) ab\bcd

Answer: (B)

Explanation: \b erases 'b', then 'c' and 'd' are printed → acd.

30. What does: printf("%c", 7); print?

```
printf("%c", 7);
```

A) 7

B) G

C) Bell sound (no visible output)

D) \a

Answer: (C)

Explanation: ASCII 7 is the bell character (same as \a). It produces a beep sound, not visible text.

31. What is the output of: printf("Line1\nLine2\nLine3\n");

```
printf("Line1\nLine2\nLine3\n");
```

A) Line1Line2Line3

B) Line1\nLine2\nLine3\n

C) Line1 Line2 Line3

D) Error

Answer: (C)

Explanation: Three \n characters each start a new line. Output: Line1, Line2, Line3 each on new lines.

32. What is the output of: printf("\n\n\n");

```
printf("\n\n\n");
```

A) Three blank lines

B) \n\n\n

C) No output

D) Three spaces

Answer: (A)

Explanation: Three \n characters produce three newlines (three blank lines).

33. char c = '\n'; printf("%d", c); What is output?

```
char c = '\n'; printf("%d", c);
```

A) 0

B) 10

C) 13

D) 92

Answer: (B)

Explanation: The ASCII value of \n is 10. Printing it as %d gives 10.

34. char c = '\t'; printf("%d", c); What is output?

```
char c = '\t'; printf("%d", c);
```

A) 8

B) 9

C) 10

D) 7

Answer: (B)

Explanation: The ASCII value of \t is 9.

35. What is the output of: printf("Hello\n\tWorld");

```
printf("Hello\n\tWorld");
```

A) Hello\n\tWorld

B) Hello World

C) HelloWorld

D) Error

Answer: (B)

Explanation: \n moves to next line, \t adds a tab. Output: Hello on one line, then tabbed World below.

36. printf("100\b\b\b200"); What is output?

```
printf("100\b\b\b200");
```

A) 100200

B) 200

C) 100\b\b\b200

D) 1200

Answer: (B)

Explanation: Three \b move cursor back 3 positions over '100', then '200' overwrites → 200.

37. What does printf("\0"); produce?

```
printf("\0");
```

A) 0

B) Null character (nothing visible)

C) Error

D) \0

Answer: (B)

Explanation: \0 is the null character (ASCII 0). It produces no visible output.

38. What is output of: printf("A\rB");

```
printf("A\rB");
```

A) AB

B) BA

C) B (overwrites A)

D) A\rB

Answer: (C)

Explanation: \r moves cursor to line start; B overwrites A. Final visible output: B.

39. printf("Sun\bMon"); -- What is output?

```
printf("Sun\bMon");
```

A) SunMon

B) SuMon

C) Sun Mon

D) Mon

Answer: (B)

Explanation: \b erases 'n', cursor goes back. Then 'M','o','n' are printed → SuMon.

40. int len = printf("OAVS"); What is the value of len?

```
int len = printf("OAVS");
```

A) 0

B) 3

C) 4

D) 5

Answer: (C)

Explanation: printf() returns the number of characters printed. 'OAVS' has 4 characters → len = 4.

41. printf("\x41"); What is output?

```
printf("\x41");
```

A) x41

B) \x41

C) A

D) 41

Answer: (C)

Explanation: \x41 is a hexadecimal escape sequence. 0x41 = 65 = ASCII 'A'.

42. printf("\101"); What is output?

```
printf("\101");
```

A) 101

B) A

C) \101

D) Error

Answer: (B)

Explanation: \101 is an octal escape sequence. Octal 101 = Decimal 65 = ASCII 'A'.

43. printf("Hi\nBye\n"); -- How many newlines are printed?

```
printf("Hi\nBye\n");
```

A) 0

B) 1

C) 2

D) 3

Answer: (C)

Explanation: There are two \n escape sequences, so two newlines are printed.

44. printf("C\tC++\tJava"); -- What separates the words?

```
printf("C\tC++\tJava");
```

A) Spaces

B) Commas

C) Horizontal tabs

D) Newlines

Answer: (C)

Explanation: \t (horizontal tab) separates the language names.

45. What is output of: printf("%s", "He said \"Hi\"");

```
printf("%s", "He said \"Hi\"");
```

A) He said Hi

B) He said \"Hi\"

C) He said "Hi"

D) Error

Answer: (C)

Explanation: \" inside a string literal prints a literal double-quote. Output: He said "Hi"

46. printf("%d", '\0'); -- What is output?

```
printf("%d", '\0');
```

A) 48

B) 0

C) 1

D) Undefined

Answer: (B)

Explanation: '\0' is the null character with ASCII value 0. Printed as integer: 0.

47. char str[] = "Hello\0World"; printf("%s", str); -- Output?

```
char str[] = "Hello\0World"; printf("%s", str);
```

A) Hello\0World

B) HelloWorld

C) Hello

D) World

Answer: (C)

Explanation: printf stops at \0 (null terminator). Only 'Hello' is printed.

48. printf("\\n"); -- What is output?

```
printf("\\n");
```

A) Newline

B) \n

C) \\n

D) n

Answer: (B)

Explanation: \\ prints a single backslash, then n is printed literally → \n (not a newline).

49. What is output of: printf("Row1\n\rRow2");

```
printf("Row1\n\rRow2");
```

A) Row1 Row2

B) Row2 Row2

C) Row1 Row2 (\r has no visual effect after \n)

D) Row2

Answer: (C)

Explanation: \n moves to next line, \r moves to start of that line, then Row2 is printed. Net effect: Row1, then Row2.

50. printf("\a\a\a"); -- What happens?

```
printf("\a\a\a");
```

A) Prints 3 'a' characters

B) Produces 3 bell/beep sounds

C) Prints \a three times

D) Nothing

Answer: (B)

Explanation: \a is the alert/bell character. Three \a produce three beep sounds (on supported systems).

51. printf("ABCDE\b\b\b12"); -- What is final output?

```
printf("ABCDE\b\b\b12");
```

A) ABCDE12

B) AB12E

C) AB12DE

D) 12CDE

Answer: (B)

Explanation: ABCDE is printed. Three \b move cursor back 3 to 'C' position. Then '1' overwrites C, '2' overwrites D → AB12E.

52. printf("Name:\t%s\nAge:\t%d", "Ravi", 25); -- What is output?

```
printf("Name:\t%s\nAge:\t%d", "Ravi", 25);
```

A) Name:Ravi Age:25

B) Name:\tRavi\nAge:\t25

C) Name: Ravi Age: 25

D) Error

Answer: (C)

Explanation: \t adds tab spacing, \n goes to new line. Output is Name: (tab)Ravi, then Age: (tab)25.

53. printf("\n"); -- How many bytes does this escape sequence occupy?

```
printf("\n");
```

A) 2 bytes

B) 1 byte

C) 0 bytes

D) 4 bytes

Answer: (B)

Explanation: Escape sequences like \n occupy exactly 1 byte in memory (ASCII value 10).

54. printf("AB\rXY"); -- What is output?

```
printf("AB\rXY");
```

A) ABXY

B) XY

C) XYAB

D) AB

Answer: (B)

Explanation: AB is printed. \r moves cursor to start of line. XY overwrites AB → XY.

55. What is output of: printf("%c%c%c", '\a', '\b', '\t');?

```
printf("%c%c%c", '\a', '\b', '\t');
```

A) \a\b\t

B) Three invisible/control characters

C) abc

D) Error

Answer: (B)

Explanation: \a is bell, \b is backspace, \t is tab -- all control characters, no visible text output.

56. printf("12345\r67"); -- What is visible on screen?

```
printf("12345\r67");
```

A) 1234567

B) 67345

C) 67

D) 12345

Answer: (B)

Explanation: 12345 is written. \r goes to line start. 67 overwrites '12' → 67345.

57. printf("Hello\nWorld\n"); -- How many lines of output?

```
printf("Hello\nWorld\n");
```

A) 1

B) 2

C) 3 (including blank line)

D) 0

Answer: (C)

Explanation: 'Hello' on line 1, 'World' on line 2, final \n creates an empty line 3. So 3 lines.

58. char ch = '\\\\'; printf("%c", ch); -- Output?

```
char ch = '\\'; printf("%c", ch);
```

A) \\

B) \

C) ch

D) Error

Answer: (B)

Explanation: \\' is a single backslash character. Printing it with %c gives \.

59. printf("I\'m fine"); -- Output?

```
printf("I\'m fine");
```

A) I\'m fine

B) I'm fine

C) Im fine

D) Compilation error

Answer: (B)

Explanation: \' prints a literal single quote. Output: I'm fine.

60. printf("%d %d", sizeof('\n'), sizeof('\t')); -- Output (in C)?

```
printf("%d %d", sizeof('\n'), sizeof('\t'));
```

A) 1 1

B) 2 2

C) 4 4

D) 1 2

Answer: (C)

Explanation: In C (not C++), character literals have type int (4 bytes on most platforms). Output: 4 4.

SECTION 3: USES AND APPLICATIONS OF ESCAPE SEQUENCES

61. Which escape sequence is used to print data in tabular form?

A) \n

B) \t

C) \r

D) \b

Answer: (B)

Explanation: \t (horizontal tab) aligns data in columns, useful for tabular output.

62. To clear the current line and reprint it, which escape sequence is most useful?

A) \n

B) \b

C) \r

D) \f

Answer: (C)

Explanation: \r (carriage return) moves cursor to the start of the current line, allowing overwrite of content.

63. Which escape sequence marks the end of a string in C?

A) \n

B) \t

C) \0

D) \e

Answer: (C)

Explanation: \0 (null character, ASCII 0) is the string terminator in C. All C strings end with \0.

64. Which escape sequence would you use to ring a bell/alert the user audibly?

A) \b

B) \r

C) \a

D) \s

Answer: (C)

Explanation: \a (alert/bell) produces an audible beep on terminals that support it.

65. To print a file path like C:\Users\file.txt, which escape sequence do you need?

A) \n

B) \\

C) \/

D) \p

Answer: (B)

Explanation: To print a backslash, you must use \\ (two backslashes). So C:\\Users\\file.txt in code.

66. Which escape sequence is commonly used to format output with newlines in C?

A) \t

B) \n

C) \b

D) \0

Answer: (B)

Explanation: \n is the most commonly used escape sequence for formatting output across multiple lines.

67. In scanf(), which escape sequence is used to skip whitespace including newlines?

A) \n

B) \t

C) A space character in the format string

D) \w

Answer: (C)

Explanation: A space in scanf's format string matches zero or more whitespace characters including \n and \t.

68. Escape sequences can be used in which of the following?

A) Only in printf()

B) Only in string literals

C) In string literals and character constants

D) Only in comments

Answer: (C)

Explanation: Escape sequences can be used in string literals (double quotes) and character constants (single quotes) in C.

69. Which escape sequence is used to print a double quote inside a string?

A) /"

B) \"

C) "

D) \q

Answer: (B)

Explanation: \" allows printing a literal double quote inside a string without ending the string.

70. What is the use of \f in C?

A) File input

B) Form feed (new page)

C) Function return

D) Float format

Answer: (B)

Explanation: \f is the form feed character (ASCII 12). It advances the printer/terminal to the next page.

71. Which escape sequence is useful for overwriting a progress bar on the same line?

A) \n

B) \r

C) \t

D) \a

Answer: (B)

Explanation: \r (carriage return) moves cursor to line start, allowing progress bar overwriting without scrolling.

72. How are escape sequences different from normal characters?

A) They take more memory

B) They represent control/special characters not typeable directly

C) They are slower to process

D) They work only in C++

Answer: (B)

Explanation: Escape sequences represent special/control characters (like newline, tab, bell) that cannot be typed directly in source code.

73. Which escape sequence is used for octal representation of a character?

A) \o followed by digits

B) \ followed by octal digits

C) \x followed by digits

D) \c followed by digits

Answer: (B)

Explanation: Octal escape sequences use a backslash followed by 1-3 octal digits: e.g., \101 for 'A'.

74. Which escape sequence is used for hexadecimal representation of a character?

A) \h

B) \x

C) \hex

D) \0x

Answer: (B)

Explanation: \x followed by hexadecimal digits represents a character by its hex ASCII value, e.g., \x41 = 'A'.

75. Which of the following correctly prints a single quote character?

A) printf("'");

B) printf("\'");

C) Both A and B

D) Neither

Answer: (C)

Explanation: A single quote can be printed directly in a double-quoted string (A) or using \' (B). Both are correct.

76. What happens when \b is used at the beginning of a string?

A) Nothing visible (cursor can't go back)

B) Deletes the first character

C) Moves to the previous line

D) Causes error

Answer: (A)

Explanation: \b at the start has no visible effect since the cursor is already at the leftmost position.

77. In C, string literals are automatically terminated with which escape sequence?

A) \n

B) \e

C) \0

D) \z

Answer: (C)

Explanation: C automatically appends \0 (null character) at the end of every string literal.

78. printf("Name\tScore\n"); -- What is the purpose of \t here?

A) Add newline between Name and Score

B) Align Name and Score in columns

C) Add backspace

D) Add bell

Answer: (B)

Explanation: \t aligns 'Name' and 'Score' as column headers in a table-like format.

79. Which escape sequence can be used to implement a simple text animation?

A) \0

B) \n

C) \r

D) \a

Answer: (C)

Explanation: \r moves cursor to line start, allowing text to be updated in place -- the basis of simple animations.

80. In C, escape sequences are resolved at which stage?

A) Runtime

B) Linking

C) Compile time

D) Preprocessing

Answer: (C)

Explanation: Escape sequences are interpreted and converted to their ASCII values during compilation (compile time).

81. What is the primary purpose of the \0 escape sequence?

A) Print zero

B) Terminate strings and mark end of data

C) Skip a character

D) Reset output

Answer: (B)

Explanation: \0 (null) is used to terminate C strings. String functions like strlen() rely on \0 to find the end of a string.

82. printf("Error!\a"); -- What extra effect does \a produce?

A) Prints 'a'

B) Bell sound to alert user

C) Moves to next line

D) No effect

Answer: (B)

Explanation: \a produces an audible alert/bell, drawing attention to the error message.

83. Which escape sequence would you use to ensure output is on a fresh line?

A) \r

B) \t

C) \n

D) \b

Answer: (C)

Explanation: \n moves the cursor to the beginning of the next line, ensuring new output starts on a fresh line.

84. To print a backslash in a string, you write:

A) \

B) \\

C) /

D) \/

Answer: (B)

Explanation: Since \ starts an escape sequence, to print a literal backslash you need \\ in the source code.

85. Which of these is the correct way to declare a newline character?

A) char c = 'n';

B) char c = \n;

C) char c = '\n';

D) char c = "\n";

Answer: (C)

Explanation: A character constant is enclosed in single quotes. The newline character is '\n'.

86. What does puts("Hello"); do compared to printf("Hello\n");?

A) They are identical

B) puts automatically adds \n, printf does not

C) printf adds \n, puts does not

D) puts uses \r instead of \n

Answer: (B)

Explanation: puts() automatically appends a newline \n after the string, whereas printf() does not.

87. In which header file is printf() declared?

A) stdlib.h

B) string.h

C) stdio.h

D) ctype.h

Answer: (C)

Explanation: printf() is declared in <stdio.h> (Standard Input/Output header).

88. What is the output of: printf("%s", ""); -- empty string?

A) Null

B) Nothing (empty output)

C) \0

D) Error

Answer: (B)

Explanation: An empty string "" contains only \0. printf prints nothing visible.

89. How many escape sequences are in: printf("\n\t\\\"\a");?

A) 3

B) 4

C) 5

D) 6

Answer: (C)

Explanation: The escape sequences are: \n, \t, \\, \", \a -- total 5 escape sequences.

90. printf("Total:\t%d\n", 100); -- What does \t do here?

A) Adds newline after 'Total:'

B) Adds tab space between 'Total:' and 100

C) Skips printing Total:

D) Causes error

Answer: (B)

Explanation: \t inserts a horizontal tab between the label 'Total:' and the value 100.

SECTION 4: ADVANCED & TRICKY QUESTIONS

91. Which of the following is a valid octal escape sequence?

A) \089

B) \777

C) \x8G

D) \99

Answer: (B)

Explanation: \777 is a valid octal escape (digits 0-7 only). \089 and \99 are invalid (digits 8,9 not octal). \x8G is invalid hex.

92. What does sizeof("\n") return in C?

A) 1

B) 2

C) 4

D) 0

Answer: (B)

Explanation: "\n" is a string literal with 1 character plus \0 terminator = 2 bytes. sizeof returns 2.

93. Which escape sequence is used in the format string of scanf to skip newline?

A) \n explicitly

B) A space ' ' in format string

C) \r

D) \0

Answer: (B)

Explanation: A space in scanf's format string matches any whitespace including \n. Common use: scanf(" %c", &c) to skip newlines.

94. char s[] = "AB\0CD"; -- What is strlen(s)?

A) 5

B) 4

C) 2

D) 6

Answer: (C)

Explanation: strlen() stops at \0. 'AB' has 2 characters before \0, so strlen = 2.

95. char s[] = "AB\0CD"; -- What is sizeof(s)?

A) 2

B) 4

C) 5

D) 6

Answer: (D)

Explanation: sizeof() counts ALL bytes including \0. s = {'A','B','\0','C','D','\0'} = 6 bytes.

96. Which statement about \r and \n is TRUE?

A) Both move cursor to next line

B) \r moves to next line, \n moves to start

C) \n moves to start of next line, \r moves to start of current line

D) They are identical

Answer: (C)

Explanation: \n (newline) = next line. \r (carriage return) = start of same line. They behave differently.

97. printf("%d", '\\'); -- What is output?

A) 2

B) 92

C) 47

D) 0

Answer: (B)

Explanation: '\\' is the backslash character. Its ASCII value is 92.

98. printf("%d", '\n'); -- What is output?

A) 92

B) 110

C) 10

D) 13

Answer: (C)

Explanation: '\n' is the newline character with ASCII value 10.

99. What is the difference between \0 and 0 in a C string context?

A) No difference

B) \0 is null char (ASCII 0), 0 is digit character (ASCII 48)

C) Both are digit zero

D) \0 is invalid

Answer: (B)

Explanation: \0 = null character (ASCII 0, string terminator). '0' = digit character (ASCII 48). Very different values.

100. printf("\x48\x65\x6C\x6C\x6F"); -- What is output?

```
printf("\x48\x65\x6C\x6C\x6F");
```

A) Hexadecimal numbers

B) Hello

C) 48656C6C6F

D) Error

Answer: (B)

Explanation: 0x48=H, 0x65=e, 0x6C=l, 0x6C=l, 0x6F=o → 'Hello'

101. Which of these correctly represents a tab character in C?

A) '\T'

B) '\t'

C) "\t"

D) Both B and C depending on context

Answer: (D)

Explanation: '\t' is used as a character constant; "\t" is used inside a string literal. Both are correct in their respective contexts.

102. What is output of printf("%d", '\a');?

```
printf("%d", '\a');
```

A) 65

B) 7

C) 97

D) 8

Answer: (B)

Explanation: '\a' is the alert/bell character with ASCII value 7.

103. Is "\9" a valid escape sequence in C?

A) Yes, octal 9

B) No, invalid (octal uses 0-7)

C) Yes, hex 9

D) Yes, decimal 9

Answer: (B)

Explanation: \9 is invalid. Octal digits are 0-7 only. \9 would be a compilation warning/error.

104. printf("\141"); -- What is output?

```
printf("\141");
```

A) 141

B) a

C) A

D) 1

Answer: (B)

Explanation: Octal 141 = Decimal 97 = ASCII 'a'.

105. printf("\x61"); -- What is output?

```
printf("\x61");
```

A) 61

B) a

C) A

D) x61

Answer: (B)

Explanation: Hex 61 = Decimal 97 = ASCII 'a'.

106. What is the maximum number of octal digits in an octal escape sequence?

A) 1

B) 2

C) 3

D) 4

Answer: (C)

Explanation: An octal escape sequence can have a maximum of 3 digits: e.g., \777.

107. Which escape sequence has no visible effect in most modern terminal output?

A) \n

B) \a

C) \t

D) \b at line start

Answer: (D)

Explanation: \b at the very start of a line has no visible effect since the cursor is already at column 0.

108. printf("\x0A"); -- What is output?

```
printf("\x0A");
```

A) A

B) 0A

C) Newline

D) Error

Answer: (C)

Explanation: 0x0A = Decimal 10 = ASCII newline (\n). Output: moves to next line.

109. What is the escape sequence for Unicode in C (C11 standard)?

A) \u followed by 4 hex digits

B) \U followed by 8 hex digits

C) Both A and B

D) Neither -- C has no Unicode escape

Answer: (C)

Explanation: C11 introduced universal character names: \uXXXX (4-digit) and \UXXXXXXXX (8-digit).

110. printf("\?"); -- What is this escape sequence?

A) Invalid

B) Literal question mark (prevents trigraph)

C) Question mark always

D) Null character

Answer: (B)

Explanation: \? is used to prevent trigraph interpretation. It represents a literal question mark.

111. What is the output: char a='\101'; printf("%c",a);?

```
char a='\101'; printf("%c",a);
```

A) 101

B) A

C) \101

D) Error

Answer: (B)

Explanation: Octal 101 = 65 = ASCII 'A'.

112. Can escape sequences be nested in C?

A) Yes

B) No

C) Only \n can be nested

D) Depends on compiler

Answer: (B)

Explanation: Escape sequences cannot be nested in C. Each starts with \ and represents exactly one character.

113. printf("\115"); -- What is the output?

```
printf("\115");
```

A) 115

B) M

C) m

D) Error

Answer: (B)

Explanation: Octal 115 = Decimal 77 = ASCII 'M'.

114. Which is the correct escape sequence for carriage return + newline (Windows line ending)?

A) \n\r

B) \r\n

C) \nr

D) \rn

Answer: (B)

Explanation: Windows line endings use \r\n (CR+LF). \r moves to line start, \n then moves to next line.

115. printf("%c", '\b'+2); -- What is output?

```
printf("%c", '\b'+2);
```

A) \b+2

B) Newline

C) Tab

D) \n

Answer: (B)

Explanation: '\b' = ASCII 8. 8+2 = 10 = ASCII '\n' (newline). Output: newline character.

116. What does \e represent in GCC (non-standard)?

A) Equal sign

B) End of string

C) Escape character (ASCII 27)

D) Error

Answer: (C)

Explanation: \e is a GCC extension representing the ESC character (ASCII 27). Not part of the C standard.

117. In C, how many escape sequences are defined in the standard?

A) 8

B) 10

C) 11

D) 12

Answer: (C)

Explanation: The C standard defines 11 escape sequences: \a, \b, \f, \n, \r, \t, \v, \\, \', \", \? (plus \0, \xhh, \ooo).

118. printf("\t" == "\t"); -- What is this comparing?

A) Tab characters

B) Pointer addresses of string literals

C) Always 1

D) Error

Answer: (B)

Explanation: String literals are compared by address (pointer comparison), not content. Result is implementation-defined.

119. What is the output: printf("%d", '\r');?

```
printf("%d", '\r');
```

A) 10

B) 13

C) 7

D) 92

Answer: (B)

Explanation: '\r' (carriage return) has ASCII value 13.

120. printf("\x7"); and printf("\a"); -- Are they equivalent?

A) No

B) Yes, both represent ASCII 7 (bell)

C) \x7 is hex, \a is octal

D) Only in C++

Answer: (B)

Explanation: \x7 = hexadecimal 7 = ASCII 7 = bell character = same as \a. They are equivalent.

SECTION 5: FULL PRACTICE TEST

121. Which escape sequence would you use to separate first and last names in output?

A) \n

B) \t or space

C) \b

D) \0

Answer: (B)

Explanation: \t (tab) or a space is commonly used to separate names in output.

122. The OAVS exam answer sheet requires printing records one per line. Which escape sequence is essential?

A) \t

B) \b

C) \n

D) \r

Answer: (C)

Explanation: \n (newline) is used to print each record on a separate line.

123. A teacher wants to print a formatted table header: 'Name Score Grade'. Which escape is best?

A) \n between each word

B) \t between each word

C) \b after each word

D) \0 at end

Answer: (B)

Explanation: \t (tab) aligns the header words in tabular columns.

124. Which escape sequence below has ASCII value 9?

A) \a

B) \b

C) \t

D) \n

Answer: (C)

Explanation: \t (horizontal tab) has ASCII value 9.

125. Which escape sequence below has ASCII value 10?

A) \r

B) \n

C) \t

D) \f

Answer: (B)

Explanation: \n (newline) has ASCII value 10.

126. Which escape sequence below has ASCII value 13?

A) \n

B) \t

C) \r

D) \b

Answer: (C)

Explanation: \r (carriage return) has ASCII value 13.

127. What is the output of: printf("*\b+");?

A) *+

B) +

C) *\b+

D) * +

Answer: (B)

Explanation: * is printed, \b moves back, + overwrites * → output is '+'.

128. A C string is always terminated by which character?

A) \n

B) \t

C) \0

D) \e

Answer: (C)

Explanation: C strings are always null-terminated with \0 (ASCII 0).

129. printf("\x4F\x41\x56\x53"); -- What is output?

```
printf("\x4F\x41\x56\x53");
```

A) OAVS

B) Hexadecimal numbers

C) \x4F\x41\x56\x53

D) Error

Answer: (A)

Explanation: 0x4F=O, 0x41=A, 0x56=V, 0x53=S → 'OAVS'

130. printf("TGT\tExam\n"); -- Count the escape sequences.

```
printf("TGT\tExam\n");
```

A) 0

B) 1

C) 2

D) 3

Answer: (C)

Explanation: There are two escape sequences: \t and \n.

131. Which is NOT a standard C escape sequence?

A) \v

B) \f

C) \e

D) \a

Answer: (C)

Explanation: \e is a GCC extension, not part of the C standard. \v, \f, \a are all standard.

132. sizeof("\t\n\r") in C -- What is the value?

```
sizeof("\t\n\r")
```

A) 3

B) 4

C) 6

D) 1

Answer: (B)

Explanation: 3 escape chars + 1 null terminator \0 = 4 bytes.

133. Which escape can represent ANY character by its ASCII value?

A) \n only

B) \xhh (hex) and \ooo (octal)

C) \a only

D) None

Answer: (B)

Explanation: Hex (\xhh) and octal (\ooo) escape sequences can represent any character by its ASCII code.

134. printf("OAVS\rTGT"); -- What is displayed?

```
printf("OAVS\rTGT");
```

A) OAVSTGT

B) TGTS

C) TGT (overwrites OAV of OAVS)

D) OAVS

Answer: (B)

Explanation: OAVS is written. \r goes to line start. TGT overwrites first 3 chars → TGTS.

135. How do you print the text: She said "Hello" in C?

A) printf("She said "Hello"");

B) printf("She said \"Hello\"");

C) printf('She said "Hello"');

D) printf("She said Hello");

Answer: (B)

Explanation: \" must be used to embed double quotes inside a string literal.

136. Which of the following printf statements will NOT compile?

A) printf("\n");

B) printf("\");

C) printf("\\");

D) printf("\t");

Answer: (B)

Explanation: printf("\"); -- a lone backslash at end of string is invalid; it tries to escape the closing quote, causing a compilation error.

137. What is the output of printf("%d", '\0' + '\n');?

```
printf("%d", '\0' + '\n');
```

A) 0

B) 10

C) 1

D) 92

Answer: (B)

Explanation: '\0' = 0, '\n' = 10. Sum = 10.

138. In C, \' is needed inside which type of literal?

A) String literals only

B) Character constants only

C) Both

D) Neither

Answer: (B)

Explanation: \' is specifically needed in character constants: char c = '\''; Inside strings, a single quote doesn't need escaping.

139. printf("C is fun!\n\tLearn C!\n"); -- How many lines of output?

```
printf("C is fun!\n\tLearn C!\n");
```

A) 1

B) 2

C) 3

D) 4

Answer: (C)

Explanation: 'C is fun!' on line 1, '\tLearn C!' on line 2, trailing \n creates empty line 3. Total: 3 lines.

140. Which escape sequence is used to move the cursor to the start of the same line?

A) \n

B) \r

C) \t

D) \f

Answer: (B)

Explanation: \r (carriage return) moves the cursor to the beginning of the current line.

SECTION 6: OAVS TGT RAPID FIRE ROUND

141. OAVS TGT Exam -- The escape sequence \n is called?

A) New Line

B) Null character

C) Numeric

D) None

Answer: (A)

Explanation: \n is called the newline character.

142. OAVS TGT -- ASCII of \t is?

A) 7

B) 8

C) 9

D) 10

Answer: (C)

Explanation: ASCII of \t (horizontal tab) is 9.

143. OAVS TGT -- \0 terminates which type of data structure in C?

A) Array

B) String

C) Pointer

D) Structure

Answer: (B)

Explanation: \0 (null character) terminates strings in C.

144. OAVS TGT -- Which escape sequence makes a beep sound?

A) \b

B) \n

C) \a

D) \r

Answer: (C)

Explanation: \a (alert) makes a beep/bell sound.

145. OAVS TGT -- To print a single backslash, we use?

A) \

B) \\

C) /

D) \/

Answer: (B)

Explanation: \\ represents a single backslash character.

146. OAVS TGT -- \r stands for?

A) Right shift

B) Return type

C) Carriage Return

D) Recursive

Answer: (C)

Explanation: \r is Carriage Return -- moves cursor to start of current line.

147. OAVS TGT -- Which escape produces horizontal spacing in output?

A) \v

B) \t

C) \n

D) \0

Answer: (B)

Explanation: \t (horizontal tab) produces horizontal spacing.

148. OAVS TGT -- How many bytes does '\n' occupy in memory?

A) 2

B) 0

C) 1

D) 4

Answer: (C)

Explanation: Although written as two characters, '\n' occupies only 1 byte in memory.

149. OAVS TGT -- printf("A\nB"); outputs A and B on?

A) Same line

B) Separate lines

C) Same line with space

D) Different files

Answer: (B)

Explanation: \n moves B to the next line. A and B appear on separate lines.

150. OAVS TGT -- Which of the following lists ONLY valid C escape sequences?

A) \n, \t, \p, \b

B) \a, \b, \f, \n, \r, \t, \v

C) \n, \t, \c, \r

D) \w, \s, \d, \n

Answer: (B)

Explanation: \a, \b, \f, \n, \r, \t, \v are all standard C escape sequences. \p, \c, \w, \s, \d are not.

COMPLETE ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	B	31	C	61	B	91	B	121	B
2	C	32	A	62	C	92	B	122	C
3	B	33	B	63	C	93	B	123	B
4	A	34	B	64	C	94	C	124	C
5	B	35	B	65	B	95	D	125	B
6	B	36	B	66	B	96	C	126	C
7	B	37	B	67	C	97	B	127	B
8	B	38	C	68	C	98	C	128	C
9	B	39	B	69	B	99	B	129	A
10	B	40	C	70	B	100	B	130	C
11	C	41	C	71	B	101	D	131	C
12	C	42	B	72	B	102	B	132	B
13	B	43	C	73	B	103	B	133	B
14	C	44	C	74	B	104	B	134	B
15	D	45	C	75	C	105	B	135	B
16	B	46	B	76	A	106	C	136	B
17	A	47	C	77	C	107	D	137	B
18	C	48	B	78	B	108	C	138	B
19	B	49	C	79	C	109	C	139	C
20	C	50	B	80	C	110	B	140	B
21	C	51	B	81	B	111	B	141	A
22	C	52	C	82	B	112	B	142	C
23	C	53	B	83	C	113	B	143	B
24	C	54	B	84	B	114	B	144	C
25	C	55	B	85	C	115	B	145	B
26	B	56	B	86	B	116	C	146	C
27	B	57	C	87	C	117	C	147	B
28	B	58	B	88	B	118	B	148	C
29	B	59	B	89	C	119	B	149	B
30	C	60	C	90	B	120	B	150	B

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OAVS TGT Computer Science Exam Prep

Escape Sequences in C -- Topic 6 of 10

This book covers ALL escape sequences tested in the OAVS TGT exam:

`\n \t \b \r \a \\ \" \' \0 \f \v`

150 Questions | 6 Sections | Full Explanations | Answer Key

Best of Luck for Your OAVS TGT Exam!